

Question				Marking details	Marks Available																	
					AO1	AO2	AO3	Total	Maths	Prac												
4	(a)	(i)		Molecules with the same {molecular / chemical} formula, but different {arrangement of atoms / structure / structural formula} (1)	1			1														
		(ii)		<table><tr><td>A</td><td>B</td></tr><tr><td>starch/ amylose/ amylopectin</td><td>cellulose</td></tr><tr><td>α- glucose</td><td>β – glucose</td></tr><tr><td colspan="2">All four boxes above correct = 2 marks 2/3 boxes correct = 1 mark 0/1 box correct = 0 marks</td></tr><tr><td>1-4 glycosidic bonds / {helical / spiral / compact} structure (1) Ignore reference to branching</td><td>each {monomer / glucose} flipped by 180° (with respect to previous one) / straight chains / reference to {hydrogen bonds between adjacent chains / microfibrils} / 1-4 glycosidic bonds (1)</td></tr><tr><td>{Carbohydrate / glucose} store (1)</td><td>cell wall / structural / rigidity / strength (1)</td></tr></table>	A	B	starch/ amylose/ amylopectin	cellulose	α- glucose	β – glucose	All four boxes above correct = 2 marks 2/3 boxes correct = 1 mark 0/1 box correct = 0 marks		1-4 glycosidic bonds / {helical / spiral / compact} structure (1) Ignore reference to branching	each {monomer / glucose} flipped by 180° (with respect to previous one) / straight chains / reference to {hydrogen bonds between adjacent chains / microfibrils} / 1-4 glycosidic bonds (1)	{Carbohydrate / glucose} store (1)	cell wall / structural / rigidity / strength (1)	4	2		6		
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		(iii)		Hydrolysis is {chemical insertion / addition} of water to break the bond (1) Maltose (1)	2			2														